

*Appendix A: Proof of Theorem 1.*

The proof is divided into two part (i) and (ii) as follows:

(i) From (24), we get

$$\begin{aligned} J &= E \sum_{k=0}^{k_f} \tilde{X}^T(k) Q \tilde{X}(k) - \gamma^2 \bar{\varepsilon}^T(k) \tilde{\varepsilon}(k) \\ &= \left\{ -\tilde{X}^T(k_f+1) P \tilde{X}(k_f+1) + \tilde{X}^T(0) P \tilde{X}(0) + \sum_{k=0}^{k_f} \left( \tilde{X}^T(k) Q \tilde{X}(k) \right. \right. \\ &\quad \left. \left. + \tilde{X}^T(k+1) P \tilde{X}(k+1) - \tilde{X}^T(k) P \tilde{X}(k) - \gamma^2 \bar{\varepsilon}^T(k) \tilde{\varepsilon}(k) \right) \right\} \end{aligned} \quad (A1)$$

Substituting (21) into (A1), we can get

$$\begin{aligned} J &= E \left\{ -\tilde{X}^T(k_f+1) P \tilde{X}(k_f+1) + \tilde{X}^T(0) P \tilde{X}(0) \right. \\ &\quad \left. + \sum_{k=0}^{k_f} \tilde{X}^T(k) Q \tilde{X}(k) + \tilde{X}^T(k) \left( \bar{A} - \bar{L} \bar{C} \right)^T P \tilde{\varepsilon}(k) \right. \\ &\quad \left. + \tilde{\varepsilon}^T(k) P \left( \bar{A} - \bar{L} \bar{C} \right) \tilde{X}(k) + \tilde{X}^T(k) \left( \bar{A} - \bar{L} \bar{C} \right) P \left( \bar{A} - \bar{L} \bar{C} \right) \tilde{X}(k) \right. \\ &\quad \left. + \bar{\varepsilon}^T(k) P \tilde{\varepsilon}(k) - \tilde{X}^T(k) P \tilde{X}(k) - \gamma^2 \bar{\varepsilon}^T(k) \tilde{\varepsilon}(k) \right\} \\ &\leq \tilde{X}^T(0) P \tilde{X}(0) + \sum_{k=0}^{k_f} \left[ \tilde{X}^T(k) \tilde{\varepsilon}^T(k) \right] \\ &\quad \times \begin{bmatrix} Q - P + \left( \bar{A} - \bar{L} \bar{C} \right)^T P \left( \bar{A} - \bar{L} \bar{C} \right) & \left( \bar{A} - \bar{L} \bar{C} \right)^T P \\ P \left( \bar{A} - \bar{L} \bar{C} \right) & P - \gamma^2 I \end{bmatrix} \begin{bmatrix} \tilde{X}(k) \\ \tilde{\varepsilon}(k) \end{bmatrix} \end{aligned} \quad (A2)$$

From the BMI in (25), we get

$$J \leq \tilde{X}^T(0) P \tilde{X}(0) \quad (A3)$$

(ii) If  $\tilde{\varepsilon}(k) \in \ell_2[0, \infty)$ , then from (24) and (A3), we get

$$E \sum_{k=0}^{k_f} \left( \tilde{X}^T(k) Q \tilde{X}(k) \right) - \gamma^2 \bar{\varepsilon}^T(k) \tilde{\varepsilon}(k) \leq \tilde{X}^T(0) P \tilde{X}(0)$$

$$\text{i.e., } E \sum_{k=0}^{k_f} \tilde{X}^T(k) Q \tilde{X}(k) \leq \tilde{X}^T(0) P \tilde{X}(0) + \gamma^2 \bar{\varepsilon}^T(k) \tilde{\varepsilon}(k).$$

Since  $\tilde{\varepsilon}(k) \in \ell_2[0, \infty)$  and  $\tilde{X}(0)$  is finite,

$$E \sum_{k=0}^{k_f} \tilde{X}^T(k) Q \tilde{X}(k) < \infty.$$

i.e.,  $\tilde{X}(k) \rightarrow 0$  in m.s.s as  $k \rightarrow \infty$ .

*Appendix B: Proof of Theorem 2.*

From (25) and  $\bar{Z} = P \bar{L}$ , we get

$$\begin{aligned} &\begin{bmatrix} Q - P + \left( \bar{A} - \bar{L} \bar{C} \right)^T P \left( \bar{A} - \bar{L} \bar{C} \right) & \left( \bar{A} - \bar{L} \bar{C} \right)^T P \\ P \left( \bar{A} - \bar{L} \bar{C} \right) & -\gamma^2 I + P \end{bmatrix} \\ &= \begin{bmatrix} Q - P + \left( \bar{A} - \bar{L} \bar{C} \right)^T P P^{-1} P \left( \bar{A} - \bar{L} \bar{C} \right) & \left( \bar{A} - \bar{L} \bar{C} \right)^T P \\ P \left( \bar{A} - \bar{L} \bar{C} \right) & -\gamma^2 I + P \end{bmatrix} \\ &= \begin{bmatrix} Q - P + \left( \bar{A} - \bar{L} \bar{C} \right) P^{-1} \left( P \bar{A} - P \bar{L} \bar{C} \right) & \left( \bar{A}^T P - \bar{C}^T \bar{L}^T P \right) \\ P \bar{A} - P \bar{L} \bar{C} & -\gamma^2 I + P \end{bmatrix} \\ &= \begin{bmatrix} Q - P + \left( P \bar{A} - \bar{Z} \bar{C} \right)^T P^{-1} \left( P \bar{A} - \bar{Z} \bar{C} \right) & \bar{A}^T P - \bar{C}^T \bar{Z}^T \\ P \bar{A} - \bar{Z} \bar{C} & -\gamma^2 I + P \end{bmatrix} \\ &\leq 0 \end{aligned}$$

By Shur Complement in Lemma 2, the above BMI is equivalent to the LMI in (28).